

PUBLISHED: 18-JUN-2017
2015.0 RANGE ROVER (LG), 303-00

ENGINE SYSTEM - GENERAL INFORMATION

CYLINDER HEAD COMPRESSION TEST

- TDV8 4.4L DIESEL (G1716425)

12.25.02	CYLINDER LEAKAGE TEST - ALL CYLINDERS	4400 CC, TDV8	2.7	USED WITHINS	
----------	---	------------------	-----	-----------------	--

SPECIAL TOOL(S)

**JLR-303-1623**Adaptor, Compression
Tester**JLR-303-1629**SP 90 Degree Elbow,
Compression Tester

**CAUTION:**

Before disconnecting or removing the components, make sure the area around the joint faces and connections are clean. Plug open connections to prevent contamination.

**NOTES:**

- Removal steps in this procedure may contain installation details.
- Some variation in the illustrations may occur, but the essential information is always correct.
- The vehicle battery must be in good condition and fully charged before carrying out this procedure.

1.

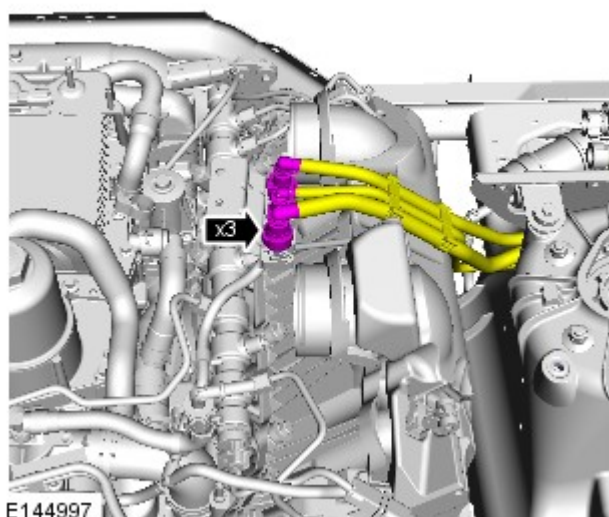
Refer to: [Glow Plugs](#) (303-07D Glow Plug System - TDV8 4.4L Diesel, Removal and Installation).

2.

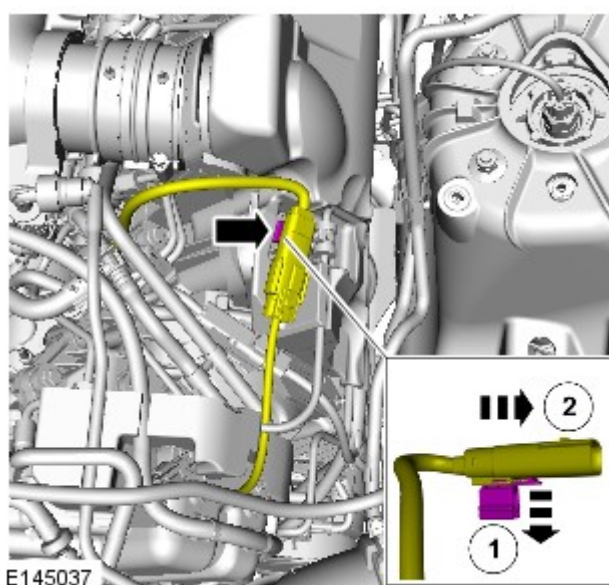
Refer to: [Left Air Cleaner](#) (303-12E Intake Air Distribution and Filtering - TDV8 4.4L Diesel, Removal and Installation).

3.**WARNING:**

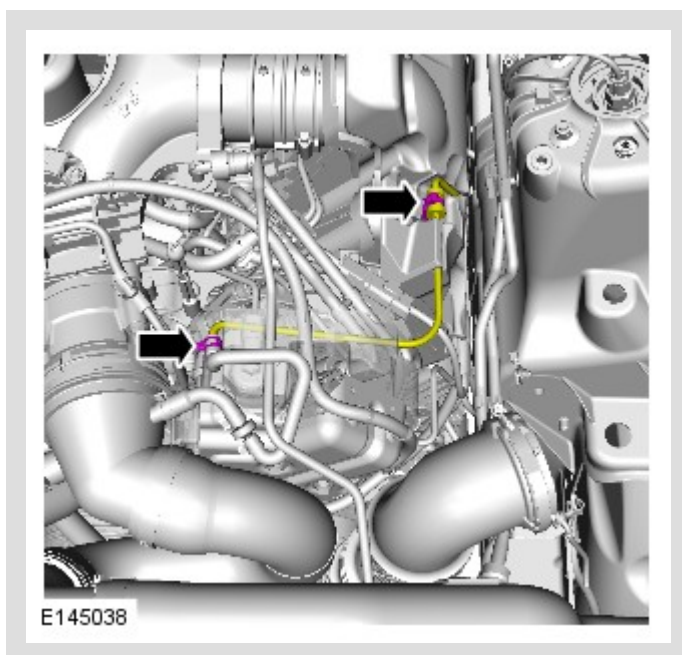
Be prepared to collect escaping fuel.



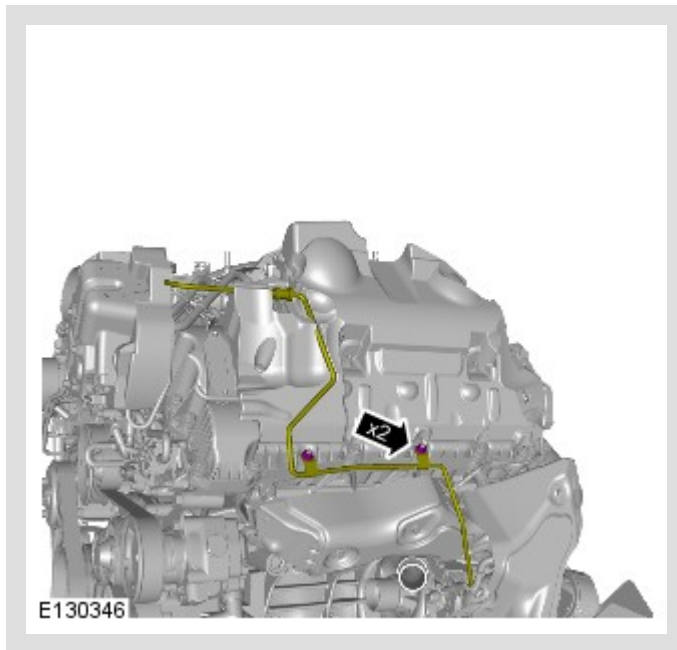
4.



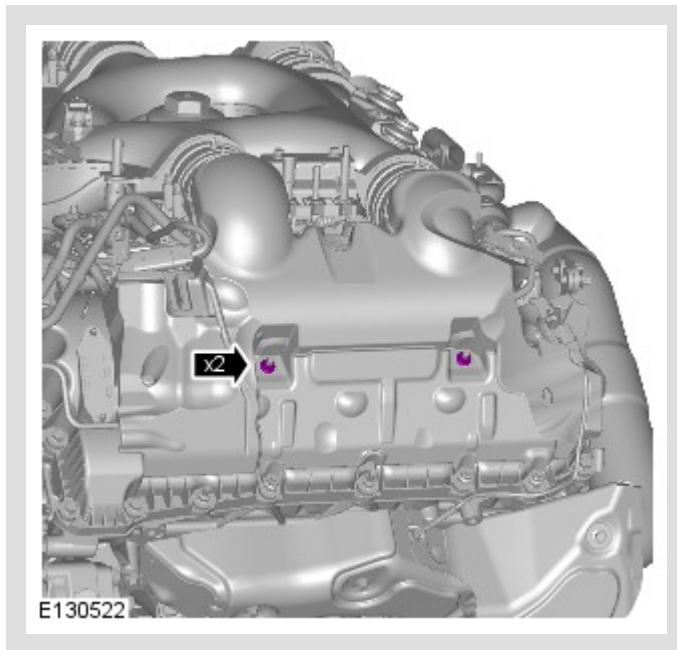
5.



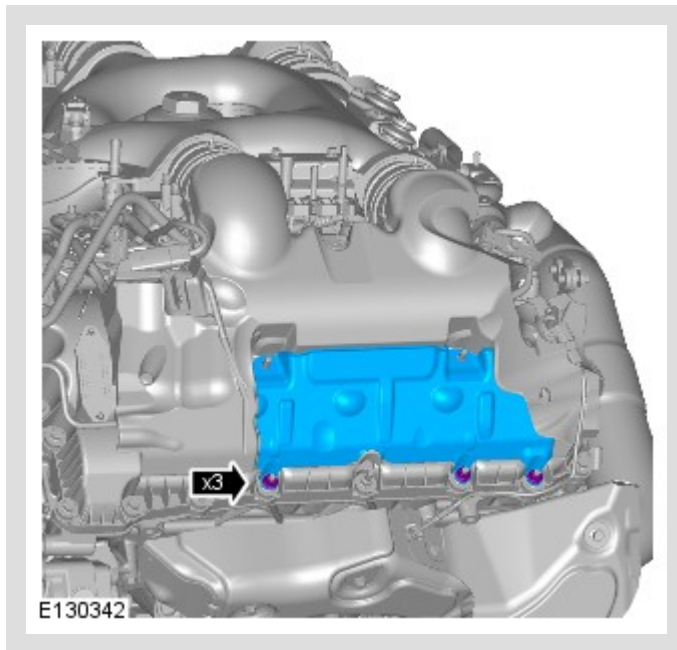
6.



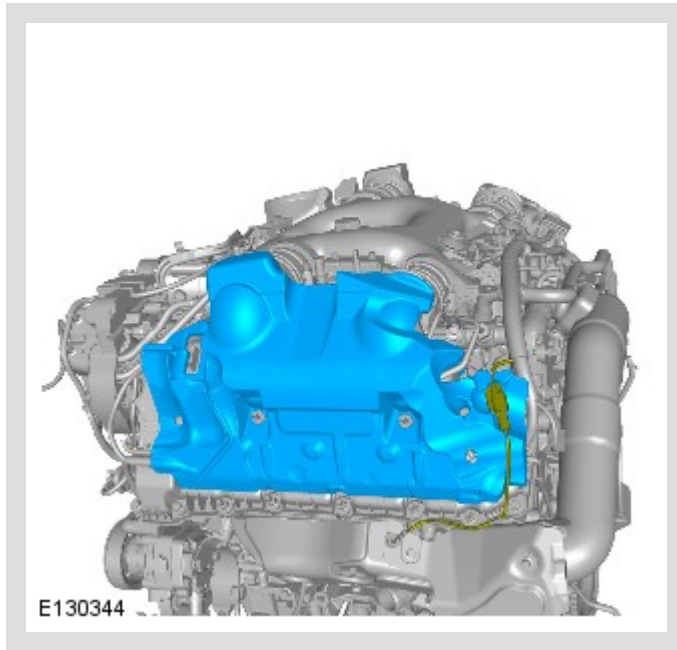
Torque: 10 Nm

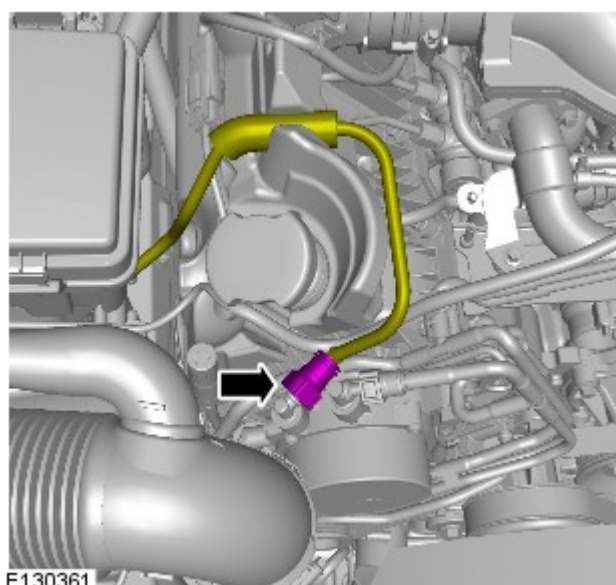


Torque: 10 Nm

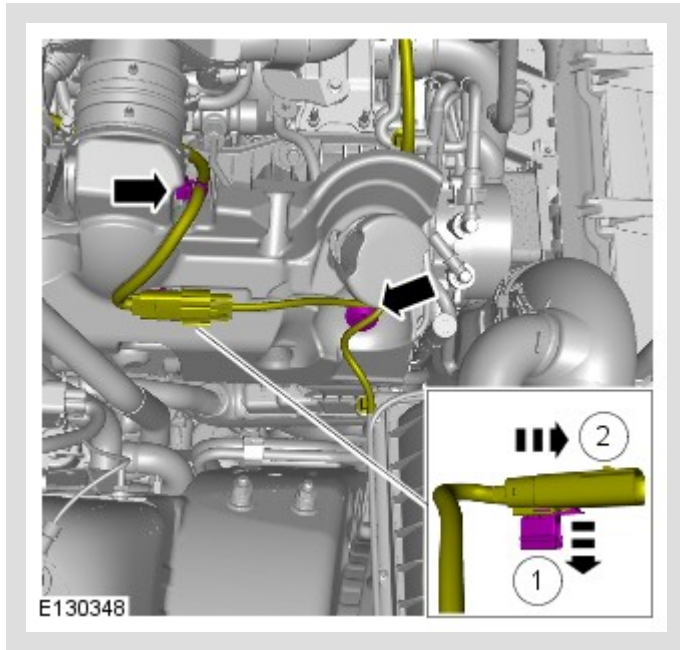


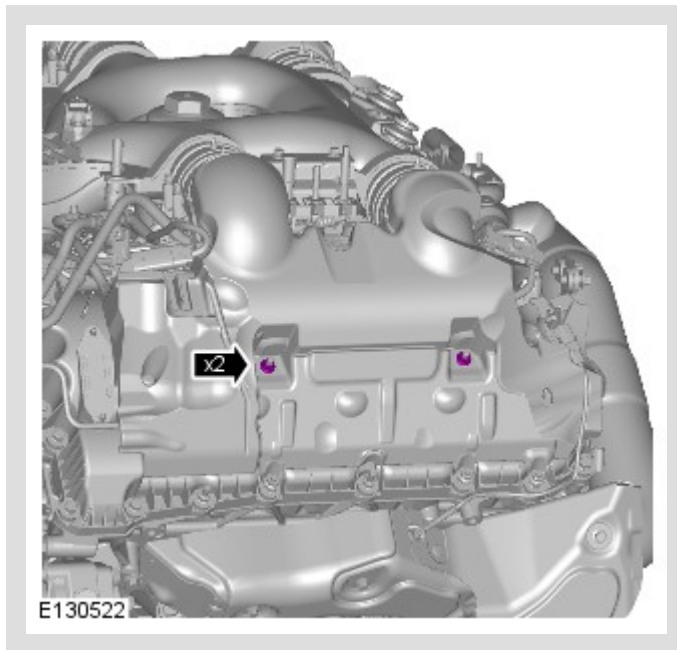
Torque: 10 Nm



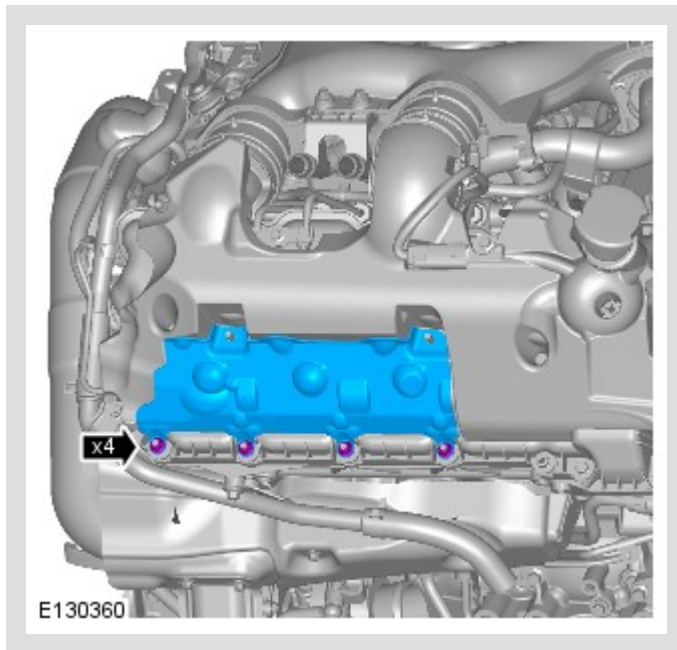


E130361

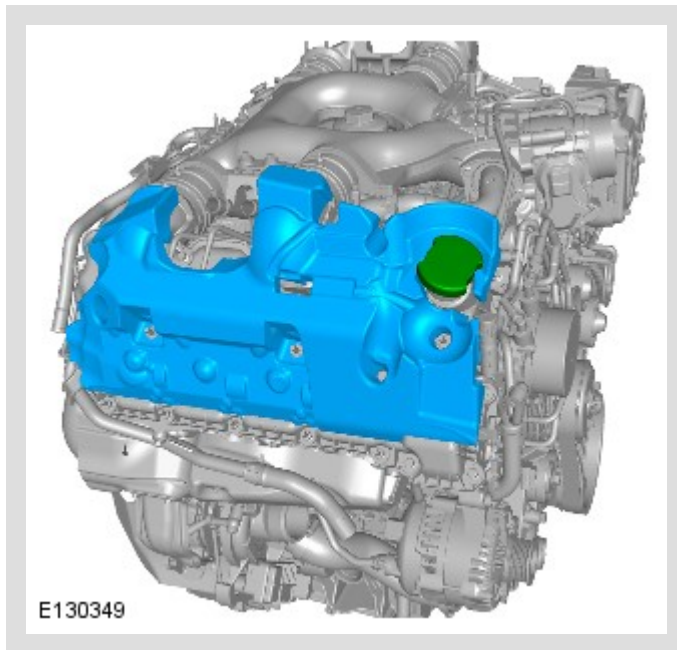




Torque: 10 Nm



Torque: 10 Nm



15.

⚠ CAUTION:

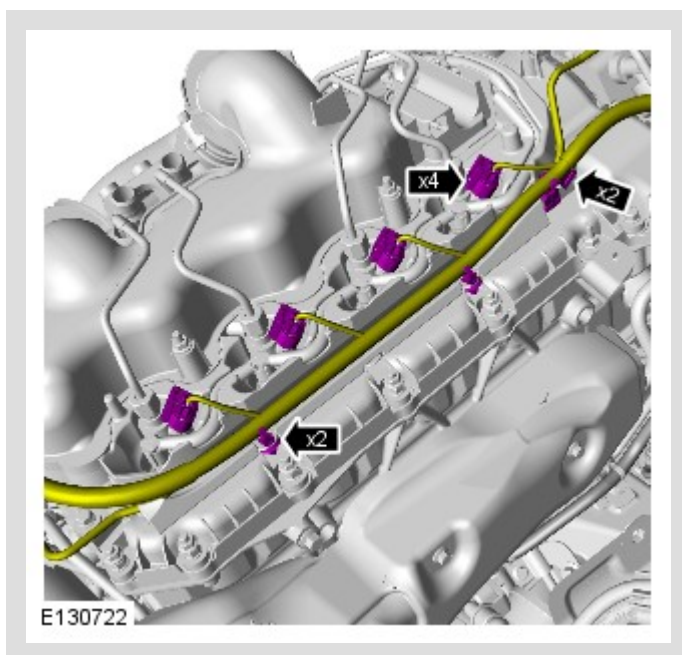
Make sure the fuel injection system is disabled before carrying out a cylinder compression test. Failure to follow this step may result in damage to the vehicle.



16.

⚠ CAUTION:

Make sure the fuel injection system is disabled before carrying out a cylinder compression test. Failure to follow this step may result in damage to the vehicle.

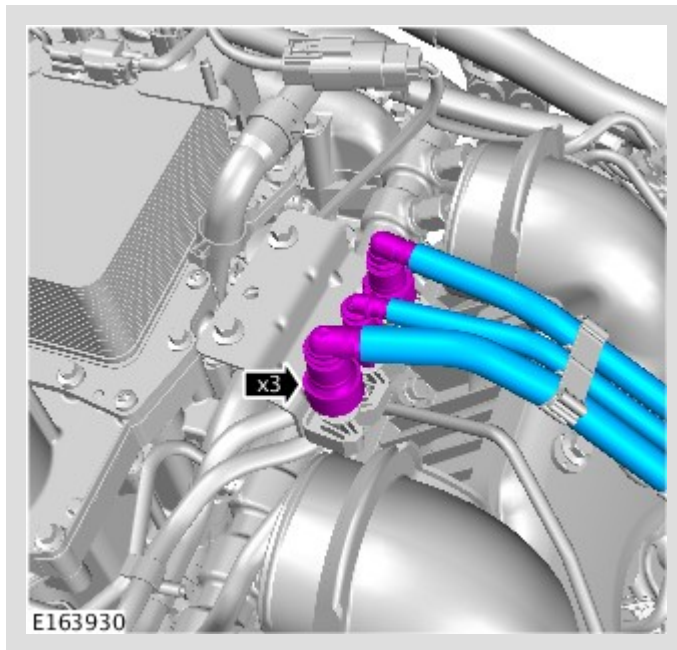


17.



CAUTION:

Install the fuel pipes.



18.

⚠ CAUTION:

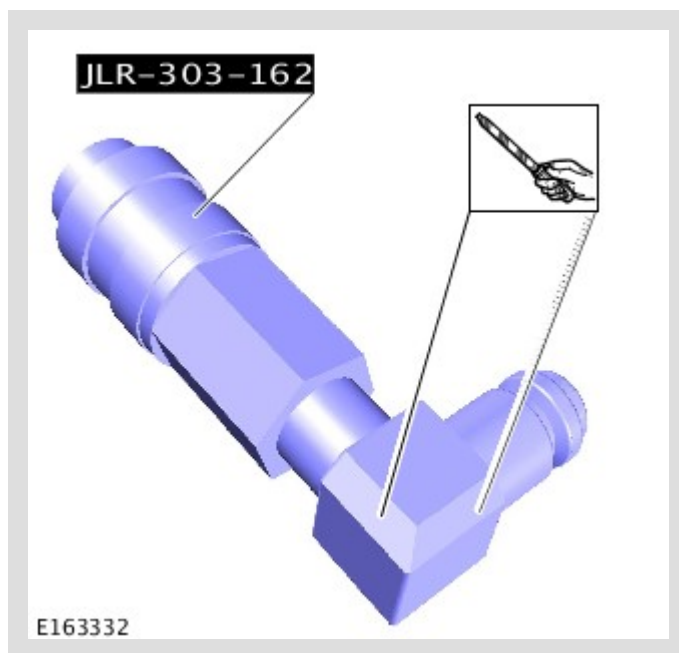
Make sure the fuel injection system is disabled before carrying out a cylinder compression test. Failure to follow this step may result in damage to the vehicle.

Crank the engine for approximately five seconds to remove any remaining fuel in the cylinders.

19.

⚠ CAUTION:

Make sure that the component is clean, free of foreign material and lubricant.



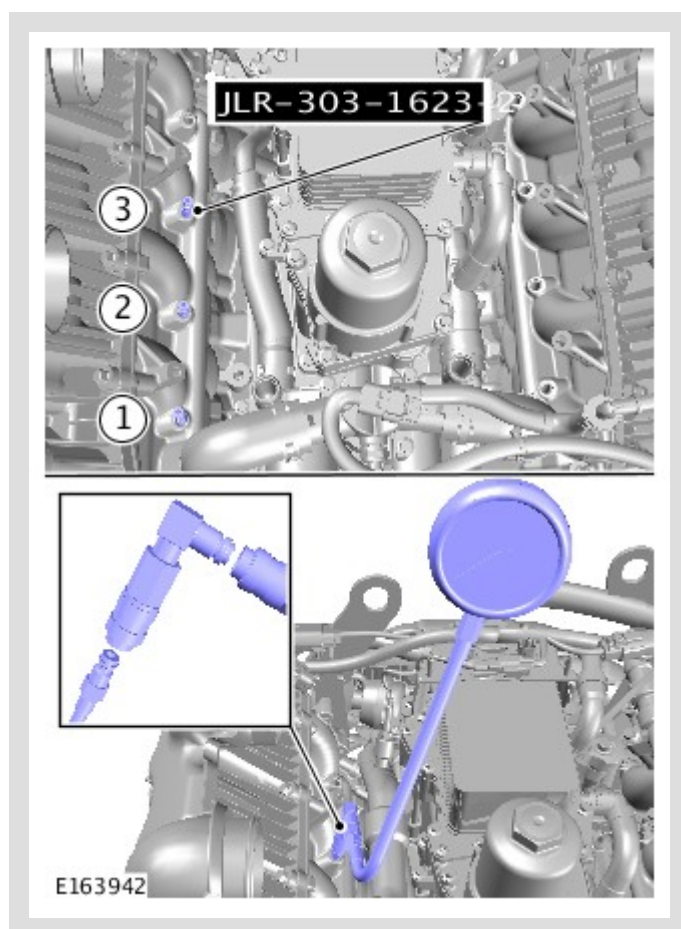
Using a suitable tool, remove the corner edges of special tool JLR-303-1629 as illustrated.

Special Tool(s): [JLR-303-1629](#)

20.

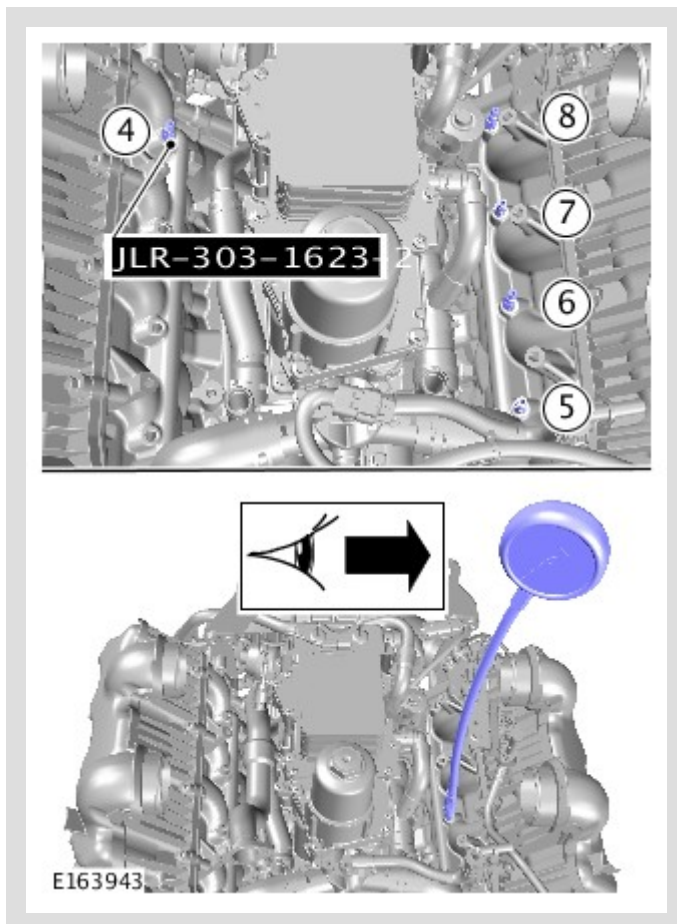
⚠ CAUTION:

The compression test gauge requires special tool JLR-303-1629 for cylinders 1,2 and 3.



Special Tool(s): [JLR-303-1623](#) , [JLR-303-1629](#)

Torque: 10 Nm



Special Tool(s): [JLR-303-1623](#)

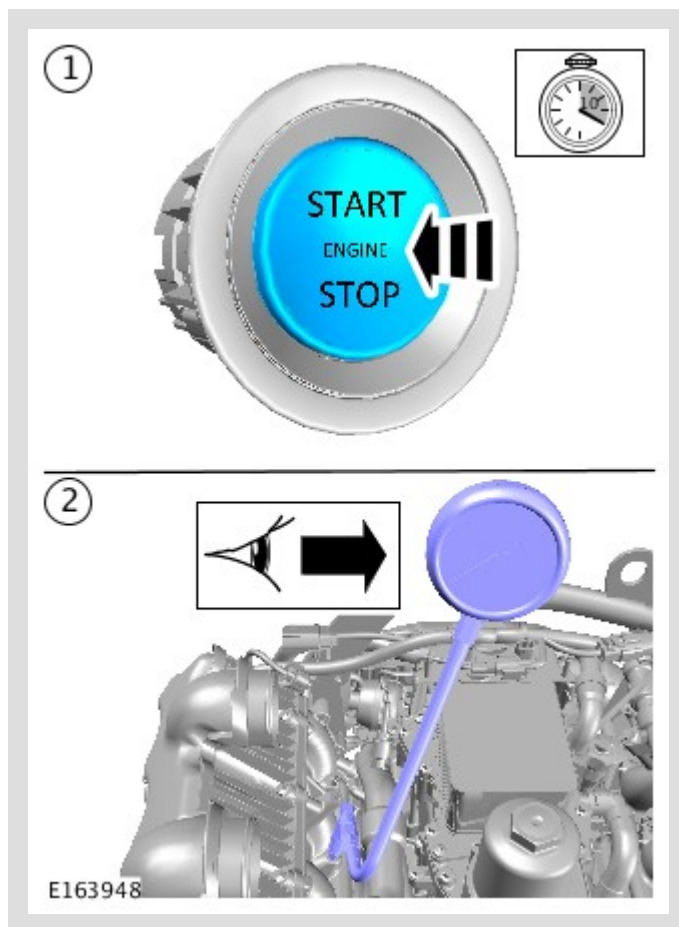
Torque: 10 Nm

22.

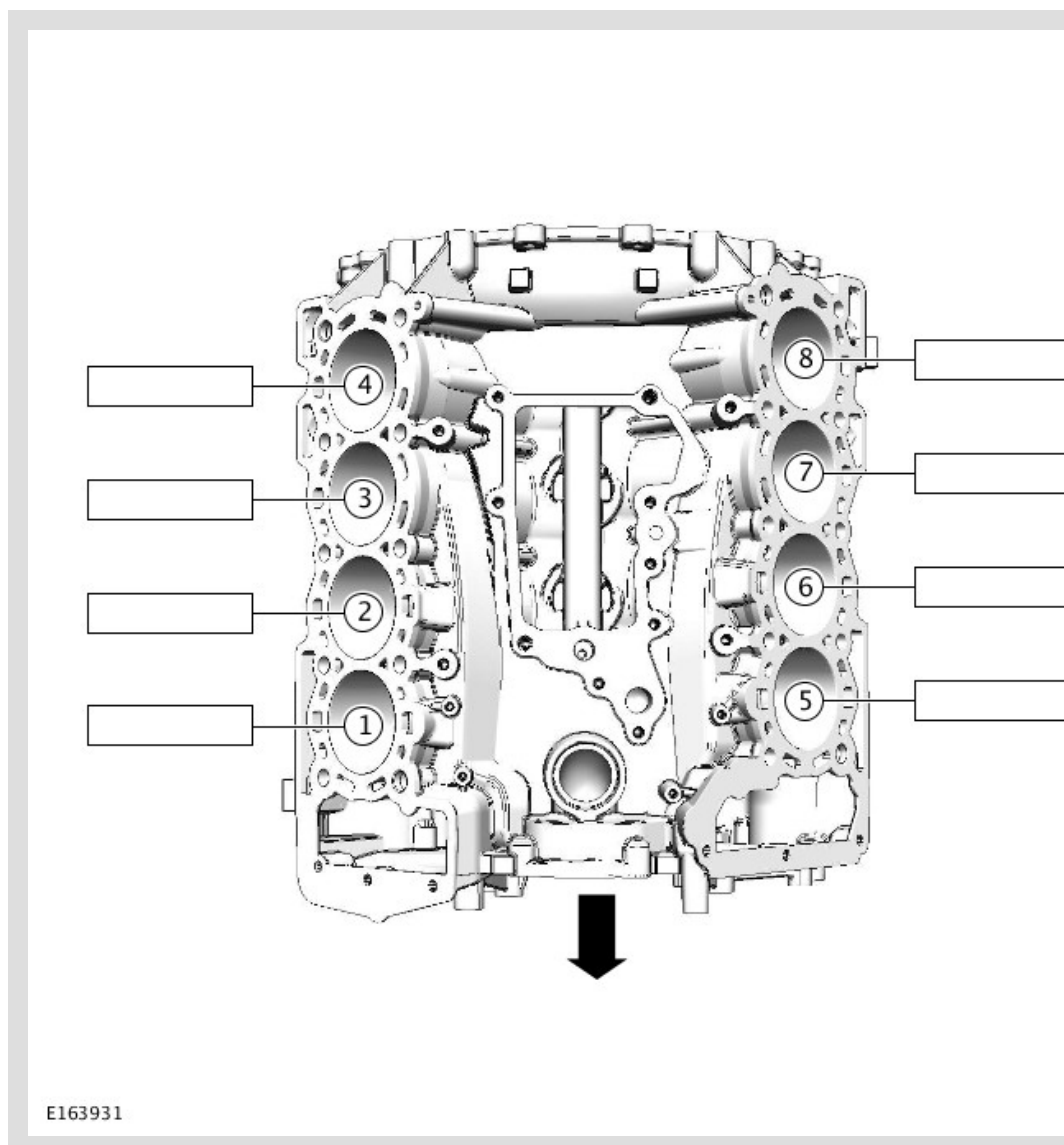


NOTES:

- This step requires the aid of another technician.
- Print graphic number E163931 in Step 23. Use this graphic to record each cylinder compression figure.
- The vehicle battery must be in good condition and fully charged before carrying out this procedure.



Crank the engine for 10 seconds and record the figure displayed on the compression test gauge. Make sure the pressure is released from the compression test gauge after each cylinder recording. Repeat the process for all cylinders.



24.



NOTE:

Remove and discard all blanking caps.

To install, reverse the removal procedure.

25. Refer to: [High-Pressure Fuel System Bleeding](#) (310-00 Fuel System - General Information, General Procedures).

26. The minimum cylinder compression reading recorded must be within 10% of the maximum cylinder compression reading recorded. If the difference across the cylinders is higher than 10% please contact dealer technical support (DTS) for further assistance.

27. Using Land Rover approved diagnostic equipment, read and clear any diagnostic trouble codes (DTCs).